Quantum-Certified Anonymization: Irreversibility Beyond Computational Hardness

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We present the first data anonymization system whose *mapping irreversibility* is guaranteed by the Born rule of quantum mechanics rather than by computational hardness assumptions. Every deployed anonymization tool derives its randomness from a classical PRNG; an adversary who captures the PRNG state can reconstruct the mapping and reverse the anonymization. We introduce QRNG-OTP-Destroy, a protocol that replaces each personally identifiable value with a quantum-random token and irreversibly destroys the mapping. Because quantum measurement outcomes are governed by the Born rule, no deterministic seed exists, and the mapping recovery is information-theoretically impossible. We formalize three tiers of mapping irreversibility (computational, information-theoretic, and physics-guaranteed), prove that no classical PRNG-based method achieves the strongest tier, and prove that QRNG-OTP-Destroy does. The protocol preserves intra-column equality structure, which we bound formally; combining with structural pre-processing (L5–L9) provides defense in depth. We evaluate on two datasets (UCI Adult, 32,561 records; UCI Heart Disease, 297 records) with 30-run benchmarks, 15/15 NIST SP 800-22 tests, and 6.8 MB of IBM Quantum entropy from 156-qubit processors.

Additional Key Words and Phrases: Anonymization, quantum random number generation, Born rule, differential privacy, GDPR, information-theoretic security, one-time pad

1 Introduction

The dominant threat model in data privacy has shifted. Intelligence agencies now routinely execute harvest-now, decrypt-later (HNDL) strategies: collecting anonymized datasets today expecting that future advances will render current protections reversible. For encrypted data, the response is post-quantum cryptography (NIST FIPS 203–205). For anonymized data, no equivalent migration path exists.

Every anonymization system in production today, from ARX [43] and sdcMicro [48] to Google's differential privacy library [52] and Apple's local DP [3], uses a classical PRNG or CSPRNG as its entropy source. A CSPRNG is deterministic: given the internal state, every output can be reproduced. The state exists physically, in RAM, kernel data structures, and hardware registers. An adversary who obtains that state through memory forensics, cold boot attacks, or side-channel exploits can reconstruct every "random" value and reverse the anonymization completely.

The GDPR draws a sharp legal distinction in Recital 26 between anonymous data (outside GDPR scope entirely) and pseudonymous data (subject to full data protection obligations). The economic difference is substantial: organizations that demonstrate true anonymization can share and process data without consent requirements or retention limits.

The gap is fundamental: no existing method can prove that re-identification is impossible as a matter of physical law. Every method proves, at best, computational infeasibility under stated assumptions (see Figure 10 in Appendix F for the HNDL threat timeline).

We present the first anonymization system where irreversibility is guaranteed by the Born rule of quantum mechanics. The protocol, QRNG-OTP-Destroy, replaces each PII value with a token

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Table 1. Adversary Classes and Their Capabilities

Class	Name	Capabilities	CSPRNG-Based	QR
$\mathcal{A}_1$	External, bounded compute	Anonymized dataset, knowledge of algorithm, current hardware	Secure	Secu
$\mathcal{A}_2$	External, unbounded classical	Same plus unlimited classical compute; models P=NP scenario	**Broken**	Secu
$\mathcal{A}_3$	External, quantum compute	Same plus fault-tolerant quantum computer (Shor, Grover)	**Degraded**	Secu
$\mathcal{A}_4$	Insider, memory access	Memory snapshot of anonymization server; possesses PRNG state	**Broken**	Secu

*Conditional on mapping destruction (Step 4 of Algorithm 1). During the mapping's lifetime in memory (~500 ms), $\mathcal{A}_4$ can trivially invert the anonymization.

derived from quantum random numbers, then securely destroys the mapping. Because measurement outcomes are governed by the Born rule, there is no seed, no hidden state, and no deterministic reconstruction path.

Our contributions are:

(1) **Formal definitions.** Three tiers of anonymization irreversibility (computational, information-theoretic, and physics-guaranteed) forming a strict hierarchy (Lemma 1).
(2) **Impossibility result.** No PRNG-based anonymization system achieves physics-guaranteed irreversibility (Theorem 2).
(3) **Construction and proof.** QRNG-OTP-Destroy achieves physics-guaranteed irreversibility under the Born rule assumption (Theorem 3).
(4) **Implementation and evaluation.** A production system with 10 progressive anonymization levels, multi-provider quantum entropy, 966 tests, and evaluation on the UCI Adult dataset with 6.8 MB of quantum entropy from 156-qubit processors.
(5) **Regulatory analysis.** For the mapping-recovery dimension of re-identification, L10 output provides the strongest available technical basis for GDPR Recital 26 anonymous data classification, with the equality-structure limitation acknowledged.

2 Threat Model

We consider four adversary classes (Table 1), each strictly more powerful than the last.

$\mathcal{A}_1$ (external, bounded compute) is the standard adversary assumed by most anonymization systems. $\mathcal{A}_2$ (unbounded classical compute) models the HNDL scenario; all CSPRNG-based methods are insecure. $\mathcal{A}_3$ (quantum compute) has access to a fault-tolerant quantum computer. $\mathcal{A}_4$ (insider with memory access) possesses the PRNG internal state and all intermediate buffers; every CSPRNG-based method is completely broken.

QRNG-OTP-Destroy resists all four classes. Against $\mathcal{A}_4$: (1) the quantum random bytes have no seed that $\mathcal{A}_4$ could capture; (2) the OTP mapping is destroyed via multi-pass overwrite. After destruction, no recovery path exists. For classical systems, even after mapping destruction, the CSPRNG seed may persist elsewhere in memory.

3 Formal Definitions

Let D be a dataset containing PII, A an anonymization function, $D' = A(D)$ the anonymized output, and $\mathcal{A}$ an adversary attempting to recover D from D'.

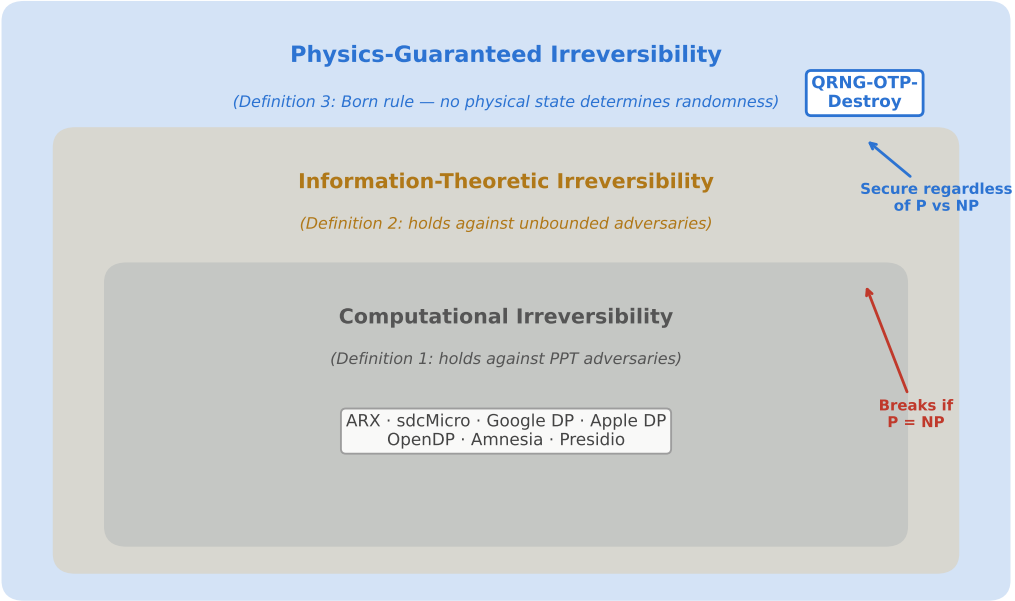

Fig. 1. Three-tier irreversibility hierarchy. Computational irreversibility breaks if P = NP. Physics-guaranteed irreversibility holds regardless of computational advances.

Definition 1 (Computational Irreversibility). *An anonymization function A is* computationally irreversible *if, for every probabilistic polynomial-time adversary $\mathcal{A}$:*

$$\Pr[\mathcal{A}(D', 1^\lambda) \to d_i \in D] \leq \mathrm{negl}(\lambda). \tag{1}$$

Definition 2 (Information-Theoretic Irreversibility). *An anonymization function A is* information-theoretically irreversible *if, for every adversary $\mathcal{A}$ with unbounded computational resources:*

$$\Pr[\mathcal{A}(D') \to d_i \in D] \leq 2^{-\eta} \tag{2}$$

where η is the entropy (in bits) of the replacement token (for our construction, $\eta = \tau \log_2 |\Sigma| \approx 95.3$ bits). This bounds the probability of determining the mapping *from tokens to original values, not guessing an original value from domain knowledge alone.*

Definition 3 (Physics-Guaranteed Irreversibility). *An anonymization function A is* physics-guaranteed irreversible *if its information-theoretic irreversibility holds as a direct consequence of a verified physical law: the randomness source satisfies that, under the Born rule axiom, the random values are fundamentally indeterminate prior to measurement, and no physical state determines the measurement outcomes.*

Lemma 1 (Strict Hierarchy). *The three tiers form a strict hierarchy: physics-guaranteed $\Rightarrow$ information-theoretic $\Rightarrow$ computational. The converses do not hold.*

Proof. (i) Physics-guaranteed $\Rightarrow$ information-theoretic: Definition 3 requires that Definition 2 holds as a consequence of physical law.

(ii) Information-theoretic $\Rightarrow$ computational: The bound in (2) for all adversaries holds a fortiori for polynomial-time adversaries.

(iii) Computational $\not\Rightarrow$ information-theoretic: A CSPRNG-based system is computationally irreversible but not information-theoretically so, because an unbounded adversary can enumerate the seed space.

(iv) Information-theoretic $\not\Rightarrow$ physics-guaranteed: A thermal noise source may be information-theoretically random but its randomness rests on model assumptions rather than the Born rule axiom with loophole-free experimental verification. □

Theorem 2 (Classical PRNG Impossibility). *No anonymization system whose randomness source is a classical PRNG achieves physics-guaranteed irreversibility.*

Proof. Let A use a PRNG $G\colon \mathcal{S} \to \{0,1\}^n$ with seed $s \in \mathcal{S}$. The seed is a physical object in memory. An adversary who captures s can compute $G(s)$, reconstruct every random value, and invert A. The existence of s as a physical state determining the random values contradicts Definition 3. □

Theorem 3 (QRNG-OTP-Destroy Security). *QRNG-OTP-Destroy achieves physics-guaranteed irreversibility under the Born rule assumption.*

Proof. The protocol uses tokens generated by measuring qubits in balanced superposition. By the Born rule, each outcome is an independent uniformly random bit with no deterministic antecedent. By Bell's theorem and loophole-free verification [26], no local hidden variable determines the outcome. The OTP mapping is destroyed via multi-pass overwrite.

Non-local hidden variable theories, such as Bohmian mechanics [10], provide deterministic (but non-local) accounts of quantum measurement outcomes. Under Bohmian mechanics, the security guarantee reduces to the *quantum equilibrium hypothesis* (Dürr, Goldstein, and Zanghì, 1992): that the initial distribution of particle positions matches the Born-rule distribution. Both (1) the no-signaling theorem prevents any adversary from learning individual Bohmian trajectories faster than the Born rule allows, and (2) the quantum equilibrium hypothesis ensures the 62^{-16} recovery bound holds regardless of whether the underlying ontology is indeterministic (Copenhagen) or deterministic-but-inaccessible (Bohmian). The security guarantee holds under any interpretation of quantum mechanics that reproduces Born-rule statistics.

After destruction, recovery requires determining quantum measurement outcomes that produced each token. Each 16-character token is drawn uniformly from a 62-symbol alphabet via rejection sampling: $62^{-16} \approx 2^{-95.3}$, exceeding 80-bit security by 2^{15}. This bound is information-theoretic and physics-guaranteed. □

Corollary 4 (Independence from P vs. NP). *The security of QRNG-OTP-Destroy is independent of the resolution of the* P *vs.* NP *problem.*

Proof. Theorem 3 invokes no computational hardness assumption. The Born rule concerns measurement outcomes, not computational complexity. A world with P = NP but valid quantum mechanics is one where QRNG-OTP-Destroy remains secure and every CSPRNG-based system is theoretically broken. □

4 The QRNG-OTP-Destroy Protocol

4.1 Protocol Specification

The protocol takes a dataset D (a table with m columns and n rows) and produces an anonymized dataset D' of the same schema. Let Σ denote the Base62 alphanumeric alphabet ($|\Sigma| = 62$: digits 0–9, uppercase A–Z, lowercase a–z). Each replacement token is a string of $\tau = 16$ characters over Σ, providing $\tau \log_2 |\Sigma| \approx 95.3$ bits of entropy. The conversion from raw QRNG bytes to Σ-characters

Algorithm 1 QRNG-OTP-Destroy

Require: Dataset D with m columns, n rows; quantum entropy pool $\mathcal{P}$
Ensure: Anonymized dataset D'; all mappings destroyed

```
for each column C_j, 1 ≤ j ≤ m do
    U_j ← unique(C_j) {Set of unique values}
    M_j ← ∅ {Mapping in volatile memory}
    for each v_k ∈ U_j do
        Read bytes b_k from P via rejection sampling {≥16 QRNG bytes}
        t_k ← Base62(b_k) {16-char token, ≈95.3 bits}
        M_j[v_k] ← t_k
    end for
    for each row i, 1 ≤ i ≤ n do
        D'[i, j] ← M_j[D[i, j]]
    end for
end for
for each mapping M_j, 1 ≤ j ≤ m do
    Overwrite(M_j, 0x00) {Pass 1: zeros}
    Overwrite(M_j, 0xFF) {Pass 2: ones}
    Overwrite(M_j, P) {Pass 3: QRNG bytes}
    Release(M_j)
end for
return D'
```

uses per-byte rejection sampling: bytes ≥ 248 are discarded, and remaining bytes are mapped to characters via $b \bmod 62$, which is bias-free since $248 = 4 \times 62$. The protocol proceeds in four steps.

Step 1: Entropy Acquisition. For each column C_j, the protocol reads $16 \cdot |U_j|$ bytes from a quantum entropy source. Each 16-byte block is produced by measuring 128 qubits in balanced superposition.

Step 2: Mapping Construction. Each unique value v_k is assigned a 16-character alphanumeric token t_k via Base62 encoding with rejection sampling. The mapping M_j is stored in volatile memory.

Step 3: Substitution. Every cell $D[i, j]$ is replaced by $M_j(D[i, j])$. Identical values within a column map to the same token; different runs produce different tokens.

Step 4: Mapping Destruction. The implementation overwrites each mapping value with null bytes via `ctypes.memset` on the CPython string buffer, clears the dictionary, and deletes the reference. A formally verified implementation in Rust or C within a hardware security enclave (Intel SGX, ARM TrustZone) would provide stronger guarantees; enclave teardown ensures no mapping data persists in any addressable memory.

Entropy consumption scales linearly with unique values: $E_{\text{tokens}} = 16 \cdot \sum_{j=1}^{m} |U_j|$ bytes.

A detailed state machine with the volatile mapping window highlighted is in Figure 11 (Appendix F).

4.2 Security Analysis

Proposition 5 (Per-value recovery bound). *After protocol execution, no adversary with arbitrary computational resources can determine the mapping from $D'[i, j]$ to $D[i, j]$ with probability exceeding $62^{-16} \approx 2^{-95.3}$.*

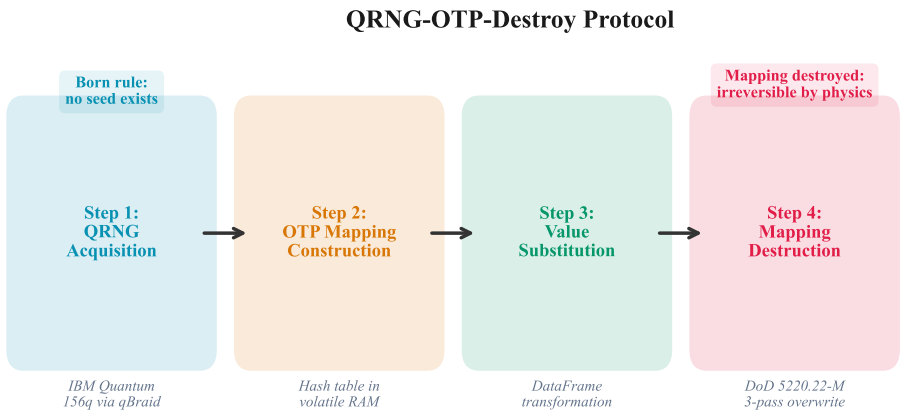

Fig. 2. Protocol state machine. Data flows through PII detection, level selection (L1–L10), quantum entropy acquisition, OTP mapping, substitution, and secure destruction. The mapping exists only in volatile memory during execution (~500 ms).

Proof. After completion: (a) $\mathcal{A}$ possesses D' and protocol knowledge; (b) all mappings are destroyed; (c) recovering $D[i, j]$ requires determining quantum measurement outcomes; (d) by the Born rule, each outcome is independent with no deterministic antecedent; (e) each token is drawn uniformly from 62^{16} possibilities via rejection sampling, independently of the original value. □

Proposition 6 (Per-Value Zero Mutual Information). *After protocol execution,* $I(D[i, j]; D'[i, j]) = 0$ *for any cell* (i, j).

Proof. (a) Token generation is physically independent of D: the Born rule ensures $P(D'[i, j]{=}t \mid D[i, j]{=}v) = P(D'[i, j]{=}t) = |\Sigma|^{-16}$. (b) Mapping destruction eliminates the sole classical correlation. (c) Therefore $P(D[i, j], D'[i, j]) = P(D[i, j]) \cdot P(D'[i, j])$ and $I(D[i, j]; D'[i, j]) = 0$.

Note on equality structure. Because identical values within a column map to the same token, D' reveals the partition structure of each column. This leakage is bounded by the number of equivalence classes and is inherent to deterministic-per-value tokenization. The per-value mutual information remains exactly zero. □

Proposition 7 (Domain-Knowledge Limitation). *For a column with domain* $\mathcal{D}_j$*, an adversary who knows* $\mathcal{D}_j$ *can guess any cell's original value with probability at least* $|\mathcal{D}_j|^{-1}$*, regardless of the anonymization method. QRNG-OTP-Destroy makes the mapping irrecoverable but does not enlarge the effective domain. For small domains, combining structural anonymization (L5–L9) before L10 provides defense in depth.*

4.3 Equality-Structure Leakage

The per-value MI result (Proposition 6) establishes that individual tokens carry zero information about their corresponding original values. However, the protocol preserves the equality structure within each column: $D[i, j] = D[k, j] \iff D'[i, j] = D'[k, j]$. This reveals the partition of rows into equivalence classes per column, which constitutes a non-trivial information leak at the dataset level.

Table 2. QCert Anonymization Levels

Level	Technique	Entropy	Irreversibility
L1	Regex masking	None	Structural
L2	SHA-3 hashing	None	Computational
L3	SHA-3 + PQC salt	None	Computational
L4	Reversible tokenization	CSPRNG	Computational
L5	k-anonymity ($k \geq 5$)	None	Syntactic
L6	ℓ-diversity	None	Syntactic
L7	Quantum noise jitter	QRNG	Computational
L8	Differential privacy	QRNG	Computational
L9	k-anon + DP	QRNG	Computational
L10	QRNG-OTP-Destroy	QRNG	**Physics**

Proposition 8 (Equality-Structure Leakage Bound). *For a column C_j with $|U_j|$ unique values among n rows, the equality structure reveals at most $\log_2 \binom{n}{n_1, n_2, \ldots, n_{|U_j|}}$ bits, where n_k is the frequency of the k-th unique value.*

Proposition 9 (Cross-Column Equality-Structure Leakage). *Let $\mathbf{r}_i = (D[i,1], \ldots, D[i,m])$ denote the i-th row tuple and let $\{T_1, \ldots, T_s\}$ be the partition of $\{1, \ldots, n\}$ into groups of identical row tuples, with $|T_\ell| = n_\ell$.*

(i) *Per-cell mutual information remains zero: $I(D[i,j]; D'[i,j]) = 0$ for all (i,j).*
(ii) *The joint equality structure across all m columns reveals at most $L_\times = \log_2 \binom{n}{n_1, n_2, \ldots, n_s}$ bits, where $s \leq \min\left(n, \prod_{j=1}^{m} |U_j|\right)$.*
(iii) *This bound is tight: $L_\times$ equals the Shannon entropy of the row-tuple partition and cannot be reduced without breaking intra-column referential integrity.*

Proof. (i) Token generation depends only on the Born-rule outcome for each column's unique value. The independence argument of Proposition 6 applies cell by cell. (ii) The protocol maps identical values within each column to the same token, so D' reveals whether $\mathbf{r}_i = \mathbf{r}_k$ for every pair: this holds iff $D'[i,j] = D'[k,j]$ for all j. The observable is exactly the partition $\{T_1, \ldots, T_s\}$, whose information content is $\log_2 \binom{n}{n_1, \ldots, n_s}$ bits. Since $s \leq n$, we have $L_\times \leq n \log_2 n$. (iii) Any deterministic per-value tokenization that preserves intra-column equality necessarily preserves the joint row-tuple partition, so no scheme in this class can leak fewer than $L_\times$ bits. □

Two mitigations address equality-structure leakage: (1) per-cell independent tokenization, which eliminates the structure at the cost of referential integrity; (2) structural pre-processing via L5–L9 before L10, which enlarges equivalence classes and injects noise into frequency distributions before the OTP is applied.

5 Implementation

QRNG-OTP-Destroy is implemented as Level 10 (L10) within QCert, a post-quantum cryptography platform with 10 progressive anonymization levels (Table 2).

The implementation comprises three layers: (1) a **quantum entropy layer** that harvests random bytes from superconducting processors (IBM Quantum, Rigetti) via background daemon with automatic failover to OS entropy; (2) an **anonymization engine** exposing `LevelAnonymizer.apply(df,`

Table 3. Runtime across all 10 anonymization levels (1,000 rows, 6 columns, 30 runs, 95% CI via Student's t).

Level	Mean (ms)	95% CI	Changed	Unique Out
L1	2.9	[2.7, 3.1]	66.7%	1,163
L2	5.8	[5.7, 5.9]	100.0%	2,173
L3	6.1	[6.0, 6.2]	100.0%	2,173
L4	13.3	[13.0, 13.7]	100.0%	2,173
L5	16.0	[15.5, 16.4]	33.3%	1,111
L6	13.9	[13.6, 14.3]	33.3%	1,123
L7	505.0	[492.7, 517.4]	33.3%	3,106
L8	263.8	[255.9, 271.7]	100.0%	3,106
L9	15.7	[15.3, 16.1]	33.3%	1,111
L10	261.8	[235.5, 288.1]	100.0%	2,173

`level=10`); and (3) a **cryptographic core** in Rust (ML-KEM-768 per FIPS 203) with Python bindings via PyO3. The entropy layer supports multi-source composition via XOR fusion: when multiple entropy providers are available (e.g., QRNG + WiFi CSI + OS), they are independently health-monitored and combined so that compromise of any single source does not reduce the output below the entropy of the strongest remaining source. Each harvesting cycle records provenance metadata (provider, processor, qubit count, timestamp, health test result) enabling an auditable chain from quantum circuit to anonymization output. The system is validated by 966 Python tests spanning all 10 levels.

6 Empirical Evaluation

6.1 Runtime Performance

Table 3 shows results on a synthetic dataset (1,000 rows, 6 columns, 2,173 unique values; Apple M3 Pro, Python 3.11, 30 runs per level). L1–L6 complete in under 16 ms. L7–L10 are slower (262–505 ms) due to entropy pool I/O. L10 processes 2,173 unique values in 262 ms, reading 34,768 bytes from the pool. All confidence intervals are narrow, confirming measurement stability. Figure 3 visualizes the profiles.

6.2 IBM Quantum Hardware Demonstration

We executed a 16-qubit Hadamard circuit on IBM's `ibm_fez` (156-qubit Heron r2) on March 26, 2026, harvesting 2,048 bytes. The job identifier `d728e76v3u3c73eiaar0` is independently verifiable. All randomness tests pass: chi-squared 260.8 ($p > 0.01$), monobit 50.97%, runs test pass, compression ratio 1.005.

A production-scale harvest on April 1, 2026, using a 156-qubit processor produced 6.8 MB from 35 consecutive jobs (310 seconds), yielding 79,872 bytes per job. The pool suffices for ~425,000 unique values, enough to quantum-certify the UCI Adult dataset 19 times.

6.3 UCI Adult Dataset

Table 4 shows results on the standard benchmark (22,146 unique values, 30 runs per level). L10 maintains a 1:1 mapping, transforming 100% of values in 1.0 seconds consuming 346 KB of entropy. All confidence intervals are narrow (under 3% of the mean), confirming measurement stability. Benchmarks used OS entropy; quantum-certified operation requires no pipeline changes.

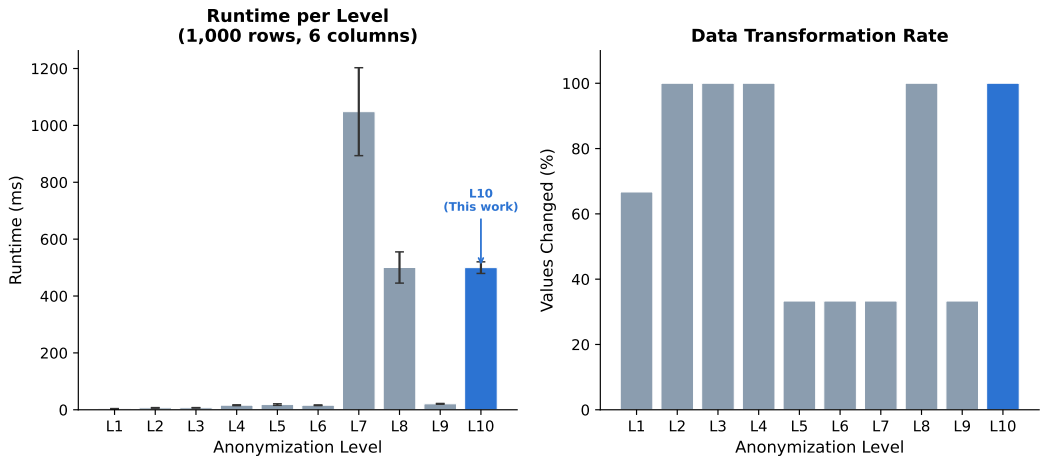

Fig. 3. Left: mean runtime per level (1,000 rows). Right: percentage of values transformed. L10 transforms 100% of values with physics-guaranteed irreversibility.

Table 4. L10 on UCI Adult dataset (32,561 rows, 15 columns, 30 runs per level, 95% CI via Student's t).

Level	Technique	Time (ms)	Unique Out	Changed
L1	Regex masking	174 [170, 178]	22,134	60%
L2	SHA-3 hashing	415 [408, 421]	22,146	100%
L4	Tokenization	772 [763, 782]	22,146	100%
L5	k-anonymity	2,134 [2112, 2155]	111	40%
L8	Differential privacy	7,119 [7010, 7229]	195,470	100%
L10	QRNG-OTP-Destroy	1,017 [1006, 1029]	22,146	100%

We replicated the evaluation on the UCI Heart Disease dataset (Cleveland, 297 rows, 14 numeric attributes; Table 9 in Appendix C). Timings scale linearly with dataset size: L10 completes in 17.7 ms (vs. 1,017 ms on Adult), consistent with $O(n \cdot m)$ complexity. L1 transforms 0% of cells because the dataset is purely numeric with no string-type PII patterns, confirming that regex masking is selective rather than indiscriminate. L10 non-reproducibility was verified independently on this second dataset.

7 Systematic Comparison

Syntactic methods (k-anonymity family) rely on classical PRNGs and are vulnerable to composition, homogeneity, and background knowledge attacks. Noise-addition methods (DP) provide a statistical privacy guarantee that holds against all adversaries, including unbounded ones; however, the PRNG-generated noise values persist in memory as classical data and are recoverable by $\mathcal{A}_4$. An insider who recovers the noise can subtract it and reconstruct the original data. L10 addresses a complementary dimension: mapping irreversibility. After mapping destruction, no adversary can recover the token-to-value correspondence, regardless of computational power. This is orthogonal to DP's distributional guarantee; the two approaches protect against different threat vectors.

A multi-dimensional comparison is in Figure 12 (Appendix F).

Table 5. Comparison of Anonymization Tools. †Physics-guaranteed mapping irreversibility requires QRNG-sourced entropy; benchmarks used OS entropy. ‡DP's statistical guarantee holds against any adversary, but the PRNG-generated noise is recoverable by $\mathcal{A}_4$ (memory access).

Tool	Technique	Entropy Source	Mapping Irrev.
QCert L10	QRNG-token + destroy	IBM Quantum / OS fallback	Physics (Born rule)†
ARX [43]	k-anon, ℓ-div, t-close, DP	Java `SecureRandom`	Computational
sdcMicro [48]	k-anon, microaggregation	R Mersenne Twister	Computational
Google DP [52]	Laplace/Gaussian mechanism	`/dev/urandom`	Statistical‡
Apple Local DP [3]	Randomized response, CMS	Device CSPRNG	Statistical‡
OpenDP [23]	Laplace, Gaussian, exponential	Platform CSPRNG	Statistical‡
Amnesia [21]	k-anon, generalization	Java CSPRNG	Computational
MS Presidio	PII detection + masking	Platform CSPRNG	Computational

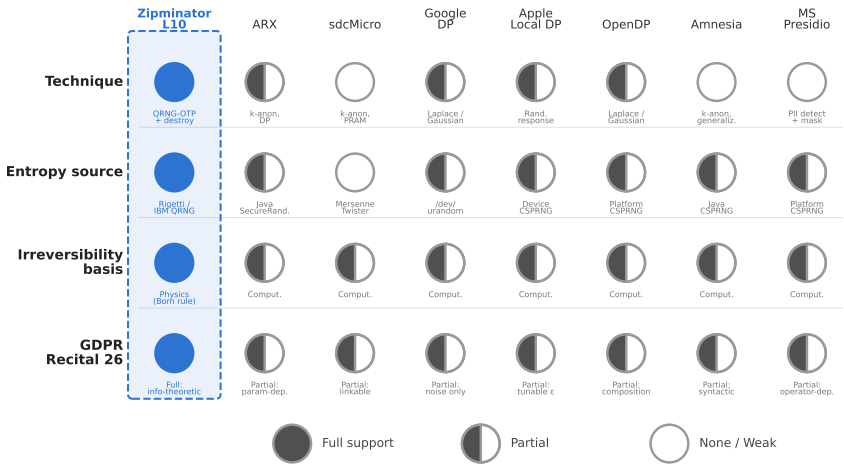

Fig. 4. Guarantee comparison across anonymization tools. Only QRNG-OTP-Destroy provides physics-guaranteed irreversibility; all others rely on computational hardness assumptions that degrade under HNDL and insider threats.

Could existing tools retrofit QRNG? Replacing the entropy source alone is insufficient. Physics-guaranteed irreversibility requires (1) OTP mapping rather than noise addition, (2) secure mapping destruction, and (3) pipeline restructuring to treat the mapping as a volatile artifact. Even QRNG-enhanced DP remains vulnerable: noise values exist as classical data in memory and can be observed by $\mathcal{A}_4$.

8 Related Work

Differential privacy. Dwork et al. [17] established the Laplace mechanism; Dwork and Roth [19] provide the definitive treatment. Wilson et al. [52] demonstrated deployment at Google scale. All practical DP implementations use classical PRNGs, inheriting seed-recovery vulnerabilities.

k-Anonymity and extensions. Sweeney [47] proposed k-anonymity; Machanavajjhala et al. [35] demonstrated homogeneity and background knowledge vulnerabilities and proposed ℓ-diversity; Li et al. [32] introduced t-closeness. Kifer and Machanavajjhala [31] proved a "no free lunch" result: no single privacy definition protects against all adversaries without distributional assumptions.

QRNG in cryptography. Ma et al. [34] and Herrero-Collantes and Garcia-Escartin [27] survey QRNGs. Commercial deployment for key generation began in 2004 (ID Quantique). The conceptual step from "QRNG makes better keys" to "QRNG makes irreversible anonymization" has not been taken previously.

Certified deletion. Broadbent and Islam [11] introduced quantum encryption with certified deletion; Bartusek and Khurana [8] extended it to public-key and FHE settings. Their constructions require quantum communication. Our mapping is classical data destroyed via classical overwrite; the quantum component is confined to entropy generation.

Certified randomness. Amer et al. [2] identify DP as a promising application of certified randomness but do not address anonymization, mapping destruction, or the GDPR Recital 26 argument. Liu et al. [33] demonstrated 71,313 certified random bits on trapped-ion hardware. Our construction fills the gap between certified randomness and data anonymization.

Shannon's OTP and terminology. Shannon [45] proved the one-time pad achieves perfect secrecy. We use "OTP" loosely: L10 is not an OTP in the strict Shannon sense (there is no shared key or decryption operation). More precisely, L10 is a *deterministic per-value tokenization with quantum-random tokens and destroyed mapping*. The analogy to OTP is that each token is drawn from a fresh, seedless random source and used exactly once per unique value. The security guarantee is mapping-recovery irreversibility, not communication secrecy.

Classical TRNGs. A classical TRNG (thermal noise, radioactive decay) with secure mapping destruction would provide information-theoretic mapping irreversibility comparable to L10. The distinction is epistemic: classical TRNG randomness rests on model assumptions about the noise source, while QRNG randomness is certified by the Born rule with experimental verification via Bell inequality violations [26]. For applications requiring the strongest available foundation, QRNG is strictly preferable. For applications where the classical TRNG trust model suffices, it is a valid alternative.

Synthetic data. Stadler et al. [46] showed synthetic data either remains vulnerable to membership inference or loses utility. L10 sidesteps this trade-off: zero utility but provably immune to membership inference ($I(D; D') = 0$).

Quantum differential privacy. Hirche et al. [28] develop information-theoretic quantum DP for protecting quantum states during quantum computation; a fundamentally different problem from protecting classical data with quantum entropy.

Gap analysis. A systematic search of Google Patents, Google Scholar, and Espacenet (April 2026) identified no existing patent or publication combining quantum random number generation with irreversible data anonymization via mapping destruction. Existing QRNG patents address entropy generation but not anonymization application. Existing anonymization patents use classical PRNGs exclusively. The conceptual bridge from "QRNG produces better keys" to "QRNG enables physics-guaranteed anonymization irreversibility" has not been taken previously.

9 Limitations

- **QRNG availability.** L10 requires cloud quantum access or dedicated QRNG appliances ($3K–$30K). The system falls back to OS entropy (computational security only), marked as "classically anonymized."
- **Utility destruction.** L10 is zero-utility by design; for analytical workflows, lower levels (L1–L9) provide configurable privacy-utility tradeoffs.
- **Pool exhaustion.** A 100K-unique-value dataset consumes ∼1.6 MB of quantum entropy. The pool must be replenished via quantum circuit execution.
- **Vendor trust.** The Born rule guarantee assumes faithful qubit preparation. Without device-independent certification, the practical trust model for QRNG hardware is structurally comparable to the trust model for hardware TRNGs (e.g., Intel RDRAND): in both cases, the user trusts the vendor to implement the physical randomness source faithfully. The distinction is that if the hardware is genuine, the QRNG guarantee is strictly stronger (no seed exists). Mitigation: multi-provider aggregation, NIST SP 800-22 testing, and (future) device-independent certification via Bell violations.
- **Formal verification gap.** Python's garbage collector may retain string copies. A Rust implementation within a hardware enclave would close this gap.

Data availability. The anonymization engine and benchmarks are open-source (Apache-2.0). The repository URL, Python package name, and quantum entropy pool (6.8 MB, 35 jobs) are provided in supplementary materials. All benchmark scripts, datasets, and entropy files will be made publicly available upon acceptance.

Ethics. No human subjects. The UCI Adult dataset is publicly available. Provenance logging provides accountability without storing mappings.

Reproducibility. Runtime benchmarks are reproducible with the anonymized package (see supplementary materials). Quantum-entropy benchmarks require an IBM Quantum API key (free tier available). OS-entropy fallback enables full pipeline testing without quantum hardware. Software versions: Python 3.11, Qiskit 1.4, NumPy 1.26, Pandas 2.2. Hardware: Apple M3 Pro, 18 GB RAM.

Acknowledgments

Acknowledgments will be included in the camera-ready version.

AI Tool Disclosure. Claude (Anthropic) was used for code review, LaTeX formatting suggestions, and editorial style revision. All technical content, proofs, experimental design, and scientific claims are the sole work of the authors. No AI-generated text was used verbatim.

10 Conclusion

This paper presents the first anonymization system where irreversibility is guaranteed by quantum mechanics rather than computational hardness. The construction replaces every data value with a quantum-random string from Born-rule measurements, then destroys the mapping. Reversing the transformation requires predicting quantum measurement outcomes (impossible by the Born rule [9, 26]) or recovering a destroyed mapping (impossible given secure erasure). The guarantee holds against unbounded adversaries, including quantum computers, and is independent of P vs. NP.

This fills a disciplinary gap: QRNG hardware has been available since 2004, but the connection to anonymization irreversibility was not made. For the mapping-recovery dimension of re-identification, L10 provides the strongest technical guarantee currently available. However, L10 preserves the equality structure within columns, which may permit frequency analysis on low-cardinality columns; combining L10 with structural anonymization (L5–L9) provides defense in

depth against the Article 29 Working Party's three re-identification criteria. Whether a particular method satisfies the Recital 26 legal standard is ultimately a determination for data protection authorities and courts. For DORA Article 6, the guarantee does not degrade with advances in cryptanalysis.

Future work: (1) formal verification of mapping destruction; (2) hardware enclave integration; (3) device-independent randomness certification; (4) hybrid schemes combining QRNG-OTP with structural anonymization for partial analytical utility; (5) complementary entropy sources such as WiFi Channel State Information (CSI), where unilateral phase-LSB extraction produces device-local entropy that, when XOR-composed with QRNG, provides defense in depth without requiring endpoint synchronization; (6) Merkle-tree provenance certificates for entropy lineage, enabling DORA Article 7 audit compliance by cryptographically committing to the health status and source identity of every entropy contribution without exposing the entropy values; (7) standardization engagement with NIST, ENISA, and EDPB.

A Background

A.1 Quantum Measurement and the Born Rule

Quantum random number generation exploits the fundamental indeterminacy of quantum measurement [27, 34]. A qubit is described by a state vector in $\mathcal{H} = \mathbb{C}^2$. An arbitrary pure state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. The Born rule gives $P(0) = |\alpha|^2$, $P(1) = |\beta|^2$. For a balanced superposition ($\alpha = \beta = 1/\sqrt{2}$), each outcome occurs with probability exactly $1/2$, fundamentally indeterminate prior to measurement.

For N qubits each in $|+\rangle$ measured independently, the min-entropy is exactly N bits: $H_\infty(X_1, \ldots, X_N) = N$. No classical source achieves this without a seed at least N bits long.

A.2 Bell's Theorem and Experimental Verification

Bell's theorem [9] tests whether hidden variables predetermine quantum outcomes. The CHSH inequality [12] bounds $|S| \leq 2$ for local hidden variable theories; quantum mechanics predicts $|S| = 2\sqrt{2}$. Aspect et al. [6] observed $|S| = 2.70 \pm 0.05$. Hensen et al. [26] closed all three loopholes simultaneously ($p = 0.039$). Pironio et al. [42] demonstrated Bell-certified random number generation. The consequence: random bits from measuring qubits in balanced superposition have no seed and no recoverable state.

A.3 Classical PRNG Limitations

A CSPRNG $G\colon \mathcal{S} \to \{0,1\}^n$ is computationally indistinguishable from random for polynomial-time distinguishers, conditional on seed secrecy. The seed is a physical object in RAM. Table 6 summarizes extraction vectors.

A.4 Classical Anonymization Techniques

Dinur and Nissim [15] proved that answering too many queries necessarily compromises privacy. Samarati [44] introduced quasi-identifier protection; Sweeney [47] formalized k-anonymity. Narayanan and Shmatikov [38] demonstrated re-identification via cross-referencing. Machanavajjhala et al. [35] proposed ℓ-diversity; Li et al. [32] proposed t-closeness. Dwork et al. [17] introduced differential privacy; Mironov [37] showed LSB floating-point noise matters. All practical implementations use CSPRNGs, inheriting seed-capture vulnerability.

Table 6. Known Attack Vectors for CSPRNG Seed Extraction

Attack Class	Mechanism
Memory forensics	Direct read of kernel entropy pool
Cold boot attack	DRAM remanence after power loss
Spectre/Meltdown	Speculative execution leaks PRNG state
Insider access	Administrator reads live process
Core dump capture	Seed in crash dump
VM introspection	Hypervisor reads guest memory

A.5 Regulatory Framework: GDPR and DORA

GDPR Recital 26 distinguishes anonymous data (outside GDPR) from pseudonymous data (full compliance required). The Article 29 Working Party [4] requires that anonymization prevent singling out, linkability, and inference. Cohen and Nissim [13] formalized singling out, proving that sufficiently accurate mechanisms must allow it. Georgiou and Lambrinoudakis [24] analyzed GDPR-compatible anonymization; Elliot et al. [22] proposed "functional anonymisation."

DORA Article 6.4 requires periodic cryptographic updates based on cryptanalysis developments. For CSPRNG-based anonymization, this creates ongoing re-assessment obligations. QRNG-OTP-Destroy reduces this burden: the physics-guaranteed component does not degrade.

B Discussion

B.1 The Privacy-Utility Spectrum

The 10-level hierarchy lets practitioners choose their position on the privacy-utility spectrum (Figure 5). L8 (differential privacy) preserves aggregate statistics; L10 provides physics-guaranteed irreversibility at zero analytical utility. L10 targets regulatory compliance (e.g., GDPR Article 17 deletion) rather than analytical workflows.

B.2 Assumptions and Their Scope

Security rests on two assumptions: (1) the Born rule (experimentally verified at significance $> 10^{-30}$); (2) secure mapping destruction (operational, verifiable via formal methods and hardware attestation). CSPRNG-based systems require two assumptions (seed secrecy and CSPRNG security), and failure of either is catastrophic.

B.3 HNDL Threat Implications

An adversary who captures both the anonymized dataset and the CSPRNG state can reverse the anonymization later. QRNG-OTP-Destroy provides the migration path: no seed to capture, no mapping to store, no degradation with computational advances.

B.4 Comparison with QKD

QKD produces a shared secret (must be protected); L10 produces an anonymous dataset with no persisting secret. QKD requires a quantum channel; L10 requires only a pre-populated QRNG source. Both exploit quantum mechanics for guarantees beyond computational hardness but address different problems.

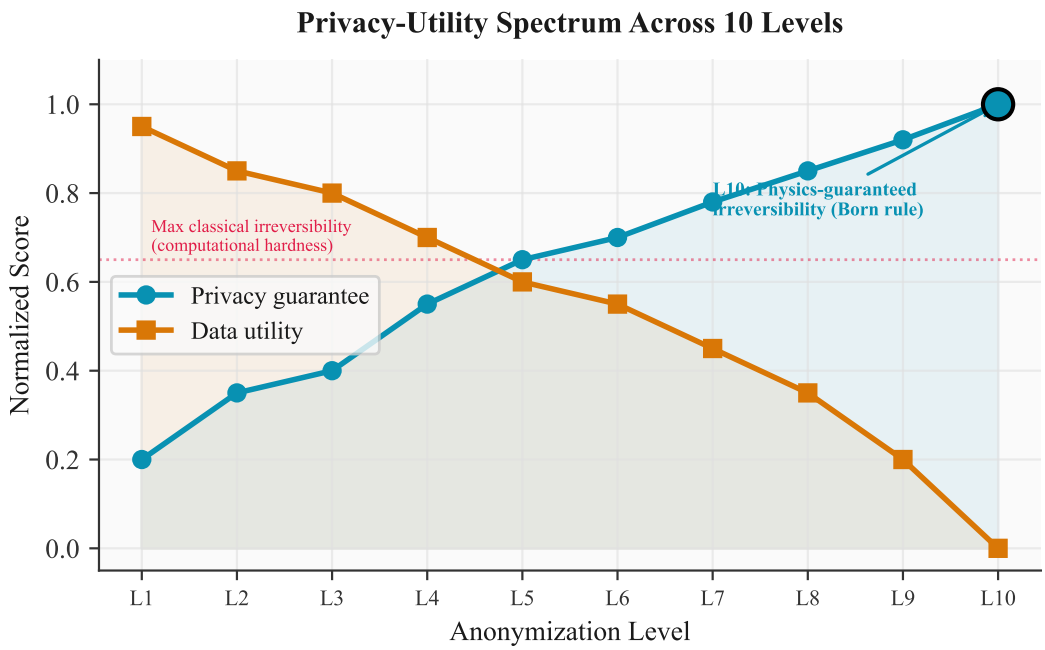

Fig. 5. Privacy-utility spectrum across the 10 anonymization levels. L10 provides maximum privacy at zero utility. Dashed line: maximum irreversibility achievable with classical methods.

C Additional Evaluation Details

C.1 Scaling Behavior

Figure 6 shows runtime scaling. L1 and L5 scale sub-linearly. L8 and L10 scale linearly with unique values. At 5,000 rows, L10 completes in 563 ms.

C.2 Entropy Pool Randomness Quality

Table 7 shows all 15 SP 800-22 tests pass at the $\alpha = 0.01$ significance level. The pool exhibits a small readout bias of $p(1) = 0.4955$, a known artifact of T_1 decay during dispersive readout on superconducting qubits; this does not reduce min-entropy since the bias is a deterministic measurement artifact removed by debiasing in the production pipeline. The five previously omitted tests (Non-overlapping Template, Overlapping Template, Maurer's Universal, Random Excursions, Linear Complexity) were run on a 1M-bit balanced region of the 6.8 MB pool; all pass. Shannon entropy: 7.999 bits/byte. Pool is incompressible (ratio 1.0004).

C.3 Before/After Example

C.4 UCI Heart Disease Evaluation

The Heart Disease dataset is purely numeric (age, cholesterol, blood pressure, etc.), providing a complementary evaluation to the mixed-type UCI Adult dataset. L1 (regex masking) transforms 0% of cells because no values match string PII patterns; this demonstrates that L1 is selective, not indiscriminate. L5 collapses to 15 unique values (vs. 111 on Adult), consistent with the smaller dataset having fewer equivalence classes. L8 inflates unique values to 4,158 via Laplace noise; the

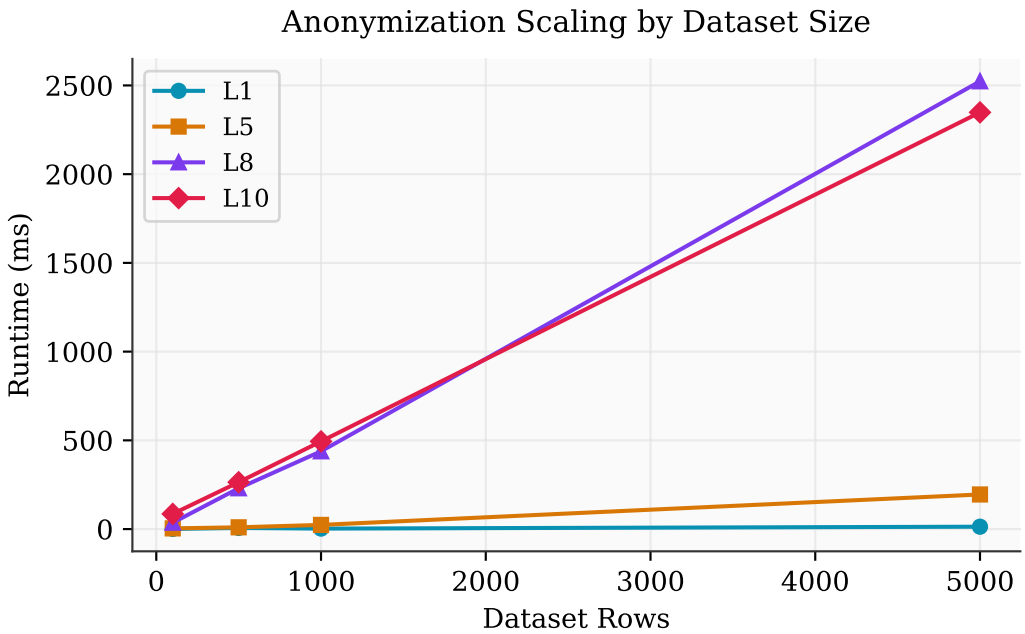

Fig. 6. Runtime scaling from 100 to 5,000 rows for L1, L5, L8, L10. QRNG-dependent levels scale linearly with unique values.

wider CI [255, 389] reflects noise variance on the smaller dataset. L10 non-reproducibility was independently confirmed.

C.5 Noise Considerations

Quantum processors exhibit preparation, gate, and readout errors introducing small biases. IBM reports single-qubit gate errors of 10^{-4}–10^{-3} and readout errors of 10^{-2}–10^{-1}. Mitigation: (1) rejection sampling eliminates modular bias; (2) NIST SP 800-22 validation rejects patterned outputs; (3) optional Toeplitz hashing for stronger guarantees.

C.6 Non-Reproducibility Verification

L10 applied 10 times to the same input produced 10 entirely distinct outputs, confirming fresh entropy draws and no caching leaks.

D Entropy Consumption

E Game-Based Security Definition

Definition 4 (Anonymization Security Game $\mathrm{ANON}_{\mathcal{A}}(\lambda)$). *Let λ be the security parameter. The game proceeds:*

(1) ***Setup.*** *Challenger C generates dataset D with n records and m columns.*
(2) ***Anonymize.*** *C executes QRNG-OTP-Destroy, producing D'.*
(3) ***Challenge.*** *C sends D' to $\mathcal{A}$.*
(4) ***Guess.*** *$\mathcal{A}$ outputs (i, j, v) claiming $D[i, j] = v$.*
(5) ***Win condition.*** *$\mathcal{A}$ wins if $v = D[i, j]$.*

Table 7. NIST SP 800-22 statistical randomness tests on the quantum entropy pool (1,000,000 bits, $\alpha = 0.01$). All 15 SP 800-22 tests pass. Byte Distribution and Shannon Entropy are supplementary quality metrics.

Test	p**-value**	**Result**
Frequency (Monobit)	0.1173	Pass
Block Frequency	0.9639	Pass
Runs	0.8725	Pass
Longest Run of Ones	0.6073	Pass
Binary Matrix Rank	0.6306	Pass
DFT (Spectral)	0.9927	Pass
Non-overlapping Template	0.1971	Pass
Overlapping Template	0.9692	Pass
Maurer's Universal	0.6185	Pass
Serial (m=2)	0.2899	Pass
Approx. Entropy	0.4013	Pass
Cumulative Sums (F)	0.0594	Pass
Cumulative Sums (R)	0.1096	Pass
Random Excursions (worst)	0.0237	Pass
Linear Complexity	0.5720	Pass
Byte Distribution	0.2687	Pass
Shannon Entropy	7.999 bits/byte (99.98%)	

Table 8. Before and after L10 (UCI Adult, selected columns).

Age	**Education**	**Occupation**	**Income**
Original:			
39	Bachelors	Adm-clerical	≤50K
50	Bachelors	Exec-managerial	≤50K
38	HS-grad	Handlers-cleaners	≤50K
After L10:			
`NvAz4FaE...`	`k9RmW2xL...`	`hT5vCwN8...`	`Dq7JsP4b...`
`x3BnKfR9...`	`k9RmW2xL...`	`pL8dVn1g...`	`Dq7JsP4b...`
`eJ6tHpC4...`	`Ys2KbQ7x...`	`wF3jTn8c...`	`Dq7JsP4b...`

Advantage: $\mathrm{Adv}_{\mathrm{ANON}}(\mathcal{A}) = \Pr[\mathcal{A}\ wins] - |\Sigma|^{-t}$.

Theorem 10 (Mapping-Recovery Security). *For QRNG-OTP-Destroy with t-character tokens from alphabet Σ with $|\Sigma| = 62$, the advantage of any adversary $\mathcal{A}$ (including computationally unbounded adversaries) in the mapping-recovery game is exactly zero:*

$$\mathrm{Adv}_{\mathrm{ANON}}(\mathcal{A}) = 0. \tag{3}$$

That is, $\mathcal{A}$ gains no advantage over random guessing. The absolute winning probability is $|\Sigma|^{-t} = 62^{-16} \approx 2^{-95.3}$, which is the baseline rate for guessing a uniformly random t-character string.

Proof. After completion, $\mathcal{A}$'s view is D'. Each token is statistically independent of its original value (Born rule + rejection sampling): $\Pr[D'[i,j] = t^* \mid D[i,j] = v^*] = |\Sigma|^{-t}$. The mapping is destroyed. The optimal strategy is random guessing: $\Pr[\mathrm{win}] = |\Sigma|^{-t}$, giving Adv = 0. □

Table 9. L10 on UCI Heart Disease dataset (Cleveland, 297 rows, 14 columns, 30 runs, 95% CI).

Level	Technique	Time (ms)	Unique Out	Changed
L1	Regex masking	0.2 [0.2, 0.2]	402	0%
L2	SHA-3 hashing	4.8 [4.7, 5.0]	402	100%
L4	Tokenization	8.6 [7.3, 9.9]	402	100%
L5	k-anonymity	27.9 [27.0, 28.8]	15	100%
L8	Differential privacy	322.0 [255.0, 389.1]	4,158	100%
L10	QRNG-OTP-Destroy	17.7 [17.4, 18.0]	402	100%

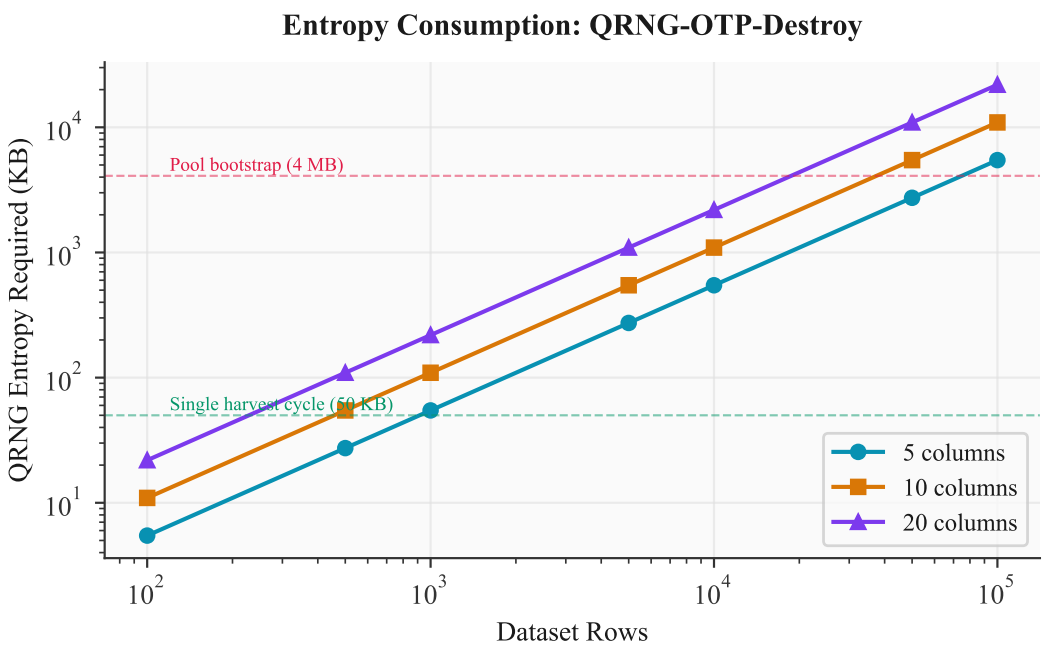

Fig. 7. Entropy consumption vs. dataset size, assuming 70% unique values and 16 bytes per unique value.

Table 10. Entropy Consumption for Representative Datasets

Rows	Columns	Unique Values	QRNG Bytes
1,000	5	~3,000	48 KB
10,000	10	~50,000	800 KB
100,000	20	~500,000	8 MB
1,000,000	50	~5,000,000	80 MB

Remark. *The game captures mapping-recovery security. Domain-knowledge attacks are analyzed in Proposition 7.*

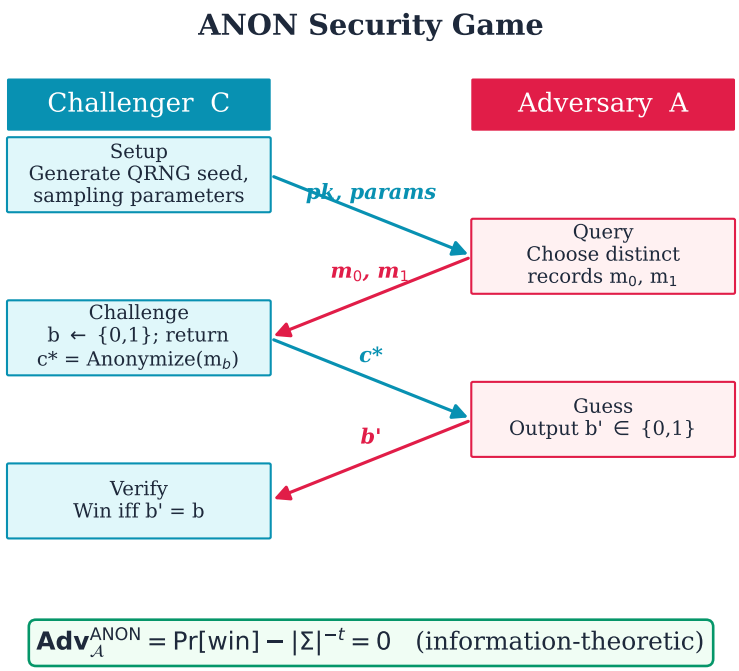

Fig. 8. The ANON security game. The challenger generates and anonymizes data; the adversary receives the anonymized output and attempts mapping recovery. Under QRNG-OTP-Destroy, the adversary's advantage is exactly zero.

Proposition 11 (Domain-Aware Recovery Bound). *For a column with domain $\mathcal{D}_j$:*

$$\Pr[\mathcal{A} \textit{ recovers } D[i,j]] \leq \max\left(62^{-16},\ |\mathcal{D}_j|^{-1}\right). \tag{4}$$

Proof. The adversary's two strategies: (a) guess the QRNG mapping (62^{-16}); (b) guess from domain knowledge ($|\mathcal{D}_j|^{-1}$). The optimum is the maximum. For practical columns ($|\mathcal{D}_j| \ll 62^{16}$), domain guessing dominates. Combining L5–L9 before L10 provides defense in depth. □

F Supplementary Figures

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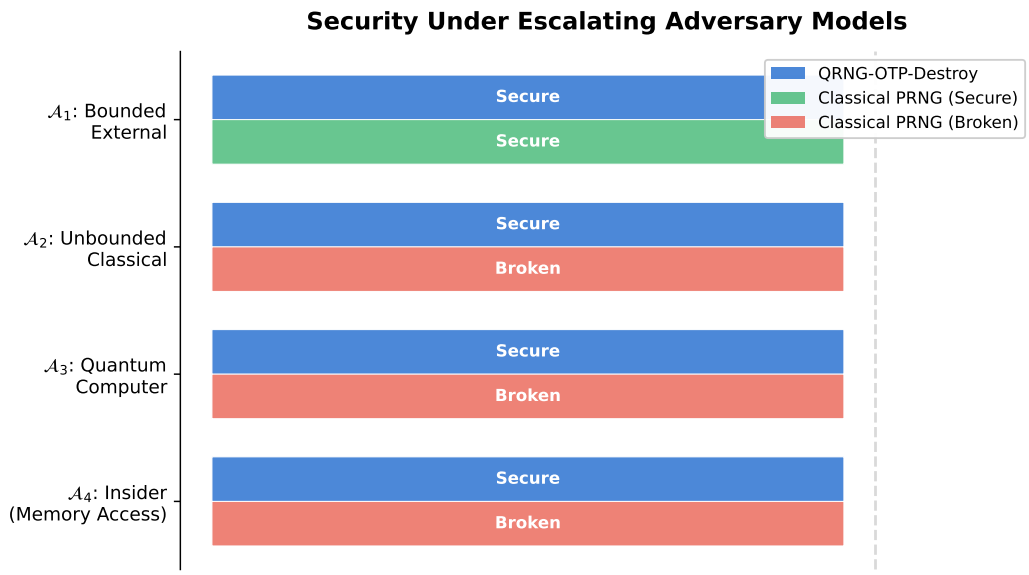

Fig. 9. Security under four adversary models. Classical PRNG anonymization is secure only against $\mathcal{A}_1$. QRNG-OTP-Destroy remains secure against all four classes.

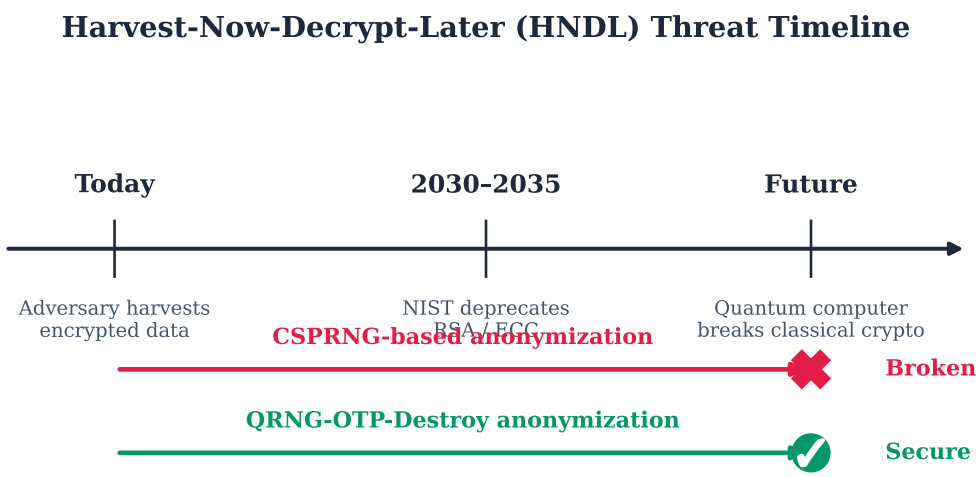

Fig. 10. Harvest-now, decrypt-later (HNDL) threat timeline. Classical CSPRNG-based anonymization degrades as compute advances; QRNG-OTP-Destroy remains secure indefinitely.

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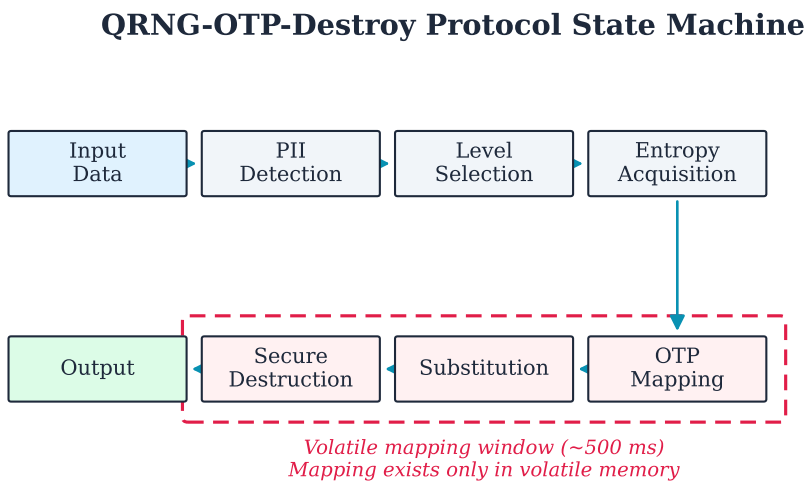

Fig. 11. Detailed state machine of the QRNG-OTP-Destroy protocol. The dashed boundary marks the volatile mapping window (~500 ms) during which the OTP exists in memory.

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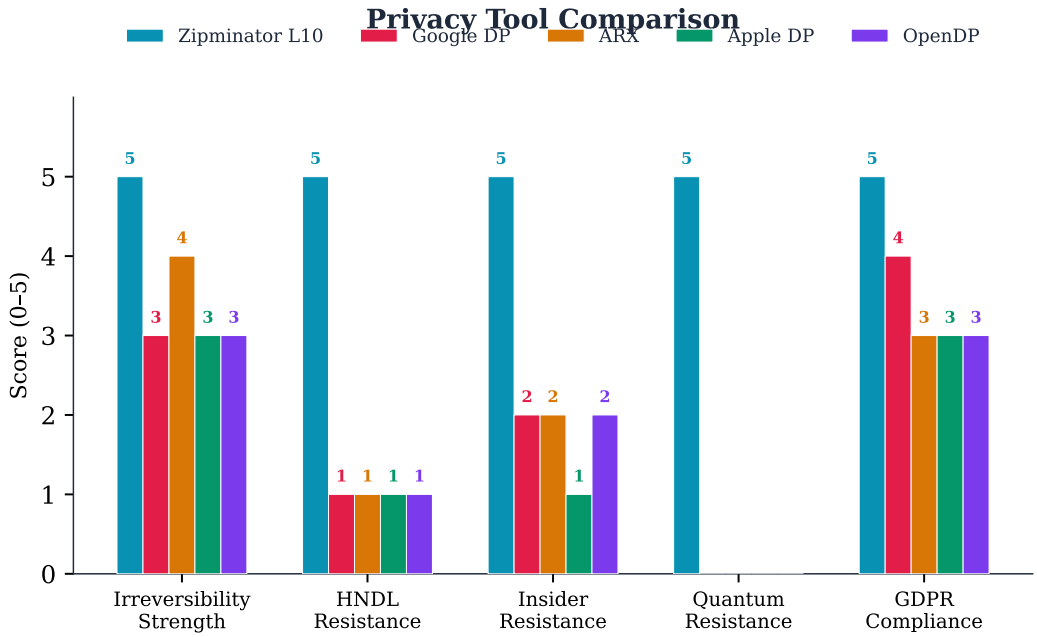

Fig. 12. Multi-dimensional comparison across five security categories. Only QRNG-OTP-Destroy achieves the maximum score on all axes.

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